

The evaluation of real-time CFD in olfactory VR

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Abstract

The sense of smell is an important sense that not only influences emotion and memory, but also plays a role in detecting danger. Complicated fluid calculations are necessary to accurately calculate odor concentration in virtual space. This research realized the interactive olfactory environment by executing fluid calculations in real time. The sensory test revealed that the direction to an odor source was easily detected when we consider influence of user's body in real-time.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Virtual reality.

Keywords

sense of smell, olfactory information, computational fluid dynamics

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1 Introduction

In virtual reality, the sense of smell has not been fully realized. However, olfaction is one of the most important human senses, as it is believed to be related to memory and emotion. We are studying the virtual environment of olfaction using real-time Computational Fluid Dynamics (CFD). Virtual environments of olfaction have been realized by several researchers. Yamada et al. used an isotropic diffusion model to generate olfactory virtual environments [Yamada et al., 2006]. In this model, odor concentration is calculated only by the distance between the user's location and the odor source. However, odor diffusion is strongly influenced by airflow. Therefore, other studies have calculated odor distribution by CFD [Matsukura et al., 2010], [Nakamoto et al., 2020]. CFD can be used to calculate more realistic odor spreading. Furthermore, previous studies have investigated the effect of the human body on odors [Uda and Nakamoto, 2024]. The human body has an effect not only as an obstacle but also as a heat source.

Virtual reality technology is expected to have significant applications in simulations. For example, a fire training simulator is not

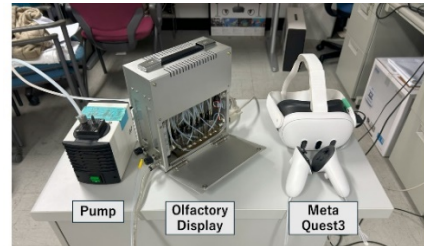


Figure 1: Devices used in experiment

only high-cost and high-risk but also require accurate olfactory information. In such cases, a more realistic olfactory experience is essential. This study investigated the usefulness of real-time CFD in the search for odor sources within a virtual environment, assuming scenarios where such odor source detection is crucial.

2 Method

This section describes the computational methods and equipment used in the sensory inspections conducted in this study.

2.1 Real-time CFD

Fluid calculations are inherently computationally demanding. Therefore, the spatial resolution is limited, and the fluid computation is performed in real time by using weakly compressible fluid computation. The lattice method was adopted because it can smoothly change the object labels in accordance with the user's movements. The program for the fluid calculations was similar to the previous study [Uda and Nakamoto, 2024].

2.2 Devices

In this study, an olfactory display using the high-speed switching of the solenoid valves [Nakamoto and Minh, 2007] was used to presenting odor to the user. This olfactory display can present 12 pre-prepared fragrances mixed with air in an arbitrary ratio. MetaQuest3 was used as the VR headset, and a PC with Intel Core i7-13700KF and NVIDIA GeForce RTX4090/24GB was used for fluid calculations and VR content control.

3 Searching the Odor Source in VR

Real-time CFD calculations make it possible to calculate the effect of user motion on odor. Since the user's body has a significant impact on the user's odor experience, it can be expected to have an impact in simulations to detect odor sources.

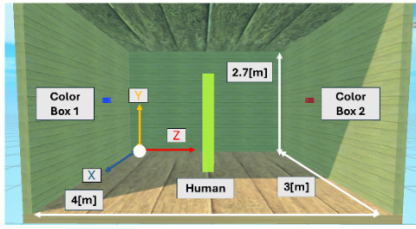


Figure 2: VR space

3.1 Setting

Small red and blue boxes were placed on both sides of a room (3.0[m]×2.7[m]×4.0[m]) in a virtual space, and a situation was prepared in which the odor drifted from one of the boxes. Grass odor (cis-3-Hexen-1-ol) was used in the experiment. To clarify the influence of the user's body, which can be realized by Real-time CFD, experiments were conducted with and without the user's body included in the calculations. The temperature of the human body was set at 32[°C] to account for the surface temperature of the skin, and the temperature of the non-human body was set at 20[°C].

In this condition, in addition to the limited size of the room, the wind velocity was set to approximately 2 [cm/s] to facilitate verification of the effect of the human body. Because it takes time for the odor to spread throughout the room, for sensory inspections, calculations are started from the results of a 20-minute simulation.

3.2 Result

Sensory testing was performed on 18 subjects. 15 subjects answered the correct odor source box when a human body was present, and 17 subjects answered the correct odor source box when no human body was present. The results showed a high percentage of correct responses both with and without a human body.

Figures 3 and 4 below show the relationship between the user's Z coordinate and the relative odor concentration at a user's nose. Note that in Figure 4, the odor source is located at a Z-coordinate of 4.0[m]. In the case with the human body model, there was almost no change in the relative odor concentration at a user's nose depending on the user's position (Figure 3). However, in the case without the human body model, the relative odor concentration at a user's nose became higher the closer the user was to the odor source (Figure 4). The user judged the direction only when he/she approached the odor source. Since the odor diffuses throughout the room, the difference in odor concentration by location of the room is small when the user's body is present. The high percentage of correct responses in the sensory test in the presence of a human body, despite the small change in the relative odor concentration along the direction to the odor source, suggests that the user used the sense of odor direction as a cue to identify the odor source even if it is away from him/her.

4 Conclusion and Future Work

In this study, we succeeded in presenting a sense of odor direction by using real-time CFD in the search for odor sources in a virtual space. The sense of odor direction realized by real-time CFD provides a more realistic odor experience and plays an important role in the

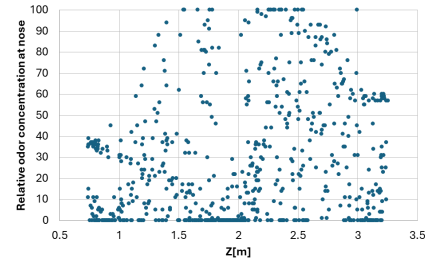


Figure 3: The relative odor concentration at a user's nose and the location (body)

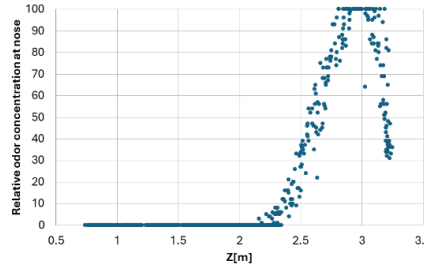


Figure 4: The relative odor concentration at a user's nose and the location (no body)

search for odor sources. On the other hand, the area used in real-time CFD is currently limited due to the highly computational cost of fluid dynamics calculations. We will adapt real-time CFD to a larger space and work on the development of odor search simulations in the future.

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